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Therefore the PR3 state with CLO power density indicator = 410/(1.79 × 18 000) = **0.01273 W.lx^{-1}.m^{-2}**.

The system power (P) calculation for the above PR5 example with CLO operation is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \qquad \text{F9}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = (8.5 + 2) = 10.5 W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power $(P) = (20 \times 10.5) + (30) = 240\,W$.

The power density indicator calculation (D_P) for the scheme for PR5 with CLO operation is—

$$D_P = \frac{P}{\sum_{i=1}^{n} \left(\bar{E}_i \cdot A_i\right)} \qquad \text{F10}$$

where

D_P = power density indicator, in watts per lux per square metre (W.lx^{-1}.m^{-2})

P = system power = 240 W

$\bar{E}_i$ = maintained average horizontal illuminance is 0.89 lx (say) (for subcategory PR5)

A_i = size of the sub-area (1000 × 18) = 18 000 m^2

n = number of sub-areas to be lit = 1

Therefore the PR5 state with CLO power density indicator = 240/(0.89 × 18 000) = **0.01498 W.lx^{-1}.m^{-2}**.

F.3 Annual energy consumption indicator (AECI)

F.3.1 General

The annual electricity consumption of a road lighting scheme depends on—

(a) the period of time for which lighting is provided;

(b) the lighting subcategory specified by the relevant lighting standard for each lighting period;

(c) the efficiency of the lighting scheme, when providing the necessary lighting for each period;

(d) the way the lighting management system follows the change in visual needs of road users; and

(e) the networked standby mode energy consumption of lighting devices during the period when the lighting is not needed (see Paragraph F2.3).

For comparison and monitoring of the energy performance of a lighting scheme, the energy consumption indicator shall take into account the annual accumulated energy use of road lighting illuminating the

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